

Total No. of Questions : 8]

SEAT No. :

PE-2525

[Total No. of Pages : 2

[6583]-51

T.E. (AI & DS)

CYBER SECURITY

(2019 Pattern) (Semester - VI)(317530)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6 and Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.

- Q1)** a) Explain Diffie-Hellman Key Exchange algorithm with example. [6]
b) Perform encryption and decryption using RSA algorithm for $p=17$, $q=11$, $e=7$ and $M=2$. [6]
c) Explain the operations of Kerberos. [6]

OR

- Q2)** a) Explain operation of MD5 message digest algorithm. [6]
b) Explain RSA algorithm with example. [6]
c) Explain X.509 Authentication Service. [6]

- Q3)** a) List and explain components of IPSec protocol. [6]
b) What is VPN? Explain components of VPN. [6]
c) Explain working of PGP in details. [6]

OR

- Q4)** a) List and explain various participants involved in Secure Electronic Transaction (SET). [6]
b) Describe the SSL protocol in details. [6]
c) What is S/MIME? What are the benefits of S/MIME? [6]

P.T.O.

- Q5)** a) Explain the firewall and its types [6]
b) What is trusted system? Explain in brief. [6]
c) Difference between IDS and IPS [5]

OR

- Q6)** a) List and explain types of Intrusion Detection System. [6]
b) What is access control security service? [6]
c) Difference between DAC and RBAC. [5]

- Q7)** a) What is cyber stalking? [6]
b) Write short note on PII. [6]
c) Explain Indian IT Act. [5]

OR

- Q8)** a) Write a short note on Botnet. [6]
b) Write short note on Cybercrime. [6]
c) Write Advantages of cyber law. [5]

